

Learning Facts at Scale with Active Reading

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Abstract—This work presents a replication study of the paper *Learning Facts at Scale with Active Reading* [1], which proposes a novel technique for improving factual recall in Large Language Models through self generated learning strategies. The original study reports substantial improvement compared to baseline synthetic data generation methods such as repetition, paraphrasing, question-answering (QA). In this project we reproduce the core Active Reading pipeline under limited computational resources, using a smaller model and reduced dataset. We compare our results against finetuned models using alternative data generation methods on two benchmarks, FinanceBench and SimpleQA. Despite the limitations of our downscaled experiments, the results support Active Reading as an effective technique for improving the internalization of factual knowledge.

Index Terms—Active Reading, Factual Recall, Self-Bleu, Learning Strategies

I. INTRODUCTION

Large Language Models encode substantial amounts of knowledge within their parameters, yet they struggle to recall long-tail facts that appear sparsely in the pretraining corpora. Simply fine-tuning or data repetition provide only limited improvement and often induce hallucinations. As a response to these challenges, Active Reading is proposed as a framework in which models actively study the corpus by generating diverse, self guided strategies that are then used to generate synthetic data. This two stage pipeline emulates human studying behaviours, like paraphrasing, active recall, analogical thinking, and proves to have better factual recall, data diversity and scalability compared to simple repetition, paraphrasing and QA.

In this work, we replicate the Active Learning framework under limited resources, using a smaller instruction model, with parameter efficient fine-tuning (QLoRA) [4], and an orders of magnitude smaller generated dataset. We compare Active Reading with the same alternative data generation methods mentioned in the paper, and provide evidence for its effectiveness even in a downscaled setting. Furthermore, we conduct a detailed analysis of our findings, assess the diversity of the different data generation methods, examine the impact of base versus instruction-tuned models, and evaluate performance on two generated datasets introduced in this study.

II. EXPERIMENTAL METHODOLOGY AND SETUP

The datasets used for synthetic data generation in this study are FinanceBench [2] and SimpleQA [3]. FinanceBench

is a question answering benchmark consisted of financial disclosures from publicly traded companies. The questions are organized in multiple categories; however for this study we focused only on information extraction questions, since the framework’s primary objective is factual recall rather than reasoning. SimpleQA includes questions whose answers can be traced in Wikipedia reference documents, provided via their urls. Both these datasets are publicly available through Hugging Face. As a preprocessing step we extracted the texts from the documents to create our corpus for synthetic generation. This resulted in 24 source documents for FinanceBench and 4020 for SimpleQA.

In order to initiate experimenting we first generated the synthetic datasets for fine-tuning using the following methods: repetition, paraphrasing, QA and Active Reading. In this replication we implemented only the task agnostic Active Reading variant, where the model generates learning strategies to study the given material generally, without conditioning on a downstream task. The task specific variant proposed in the original work was not explored due to limited time and computational resources. The generation was conducted with the use of the instruction tuned Qwen3-4B model in a 4 bit quantized configuration, instead of Llama 3.1-8B which was used in the paper. Despite having fewer parameters, Qwen3-4B has been reported to achieve competitive performance on several benchmarks, motivating its selection as a resource-efficient alternative to Llama 3.1-8B. Appropriate prompting templates were used to guide the model in producing method specific synthetic data from the original document corpus. An example of the prompt used for Active Reading strategy generation is shown in Fig. 1. Due to limited time and resources, it was not possible to process the entire corpora; consequently using only a subset of 19 documents from FinanceBench and 856 from SimpleQA for each method. During post generation analysis we observed a substantial imbalance in the number of generated tokens across methods, with paraphrasing yielding considerably fewer tokens than QA and Active Reading. To ensure equal and fair evaluation we subsampled the QA and Active Reading corpora in order to match the token count from paraphrasing. The exact procedure was applied also to the repetition dataset, which was subsampled from the original documents. While necessary for consistency, this normalization introduced information loss and represents a key limitation of the replication. The total synthetic data scale

is limited to approximately 1.5 million tokens for SimpleQA and 1.25 for FinanceBench, which are extremely below the 4 billion tokens used in the paper.

```
Consider the following document. What
are some strategies specific to this
document that I can use to help me
learn and remember all of the
information contained? Use markdown
and prefix each strategy with ##.

<document>
{chunk}
</document>
```

Fig. 1. Active Reading strategy-generation prompt.

Due to limited GPU memory, we adopted QLoRA efficient fine-tuning instead of the full scale fine-tuning procedure of the paper. The same fine-tuning configuration was applied consistently across all synthetic datasets to ensure fair comparison between data generation methods. We used a sequence length of 1024 and trained with a batch size of 16, employing sequence packing to improve computational efficiency. The maximum number of training steps was set to 330 for FinanceBench and 400 for SimpleQA, allowing the model to observe each dataset approximately three times, in alignment with the training exposure used in the original study, albeit at a substantially reduced scale. The training followed a causal language modeling objective (next token prediction) with a constant learning rate of $2e-4$. Additionally, we incorporated segments of the original corpus into the fine-tuning data using a 30–70 mixing ratio between original and synthetic data, mirroring the data mixing strategy described in the original work.

The evaluation of the data generation methods was performed by prompting the original Qwen3-4B model to judge the correctness of the predicted answers the finetuned models gave to factual questions retrieved by the HuggingFace datasets, compared to the ground truth. We further explored performance when the models were provided with the relevant documents containing the answer to each question, which we treat as a gold-context (ceiling) baseline. The complete pipeline is shown in Figure 2.

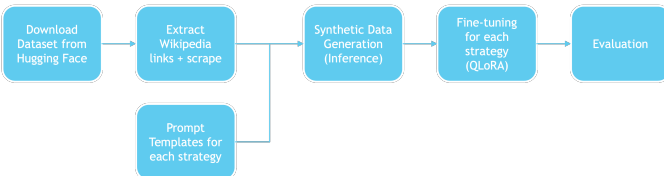


Fig. 2. Pipeline consisting of data extraction, synthetic data generation, fine-tuning and evaluation.

We further expanded our experiments by analyzing the effect of using base versus instruction tuned Qwen models.

In addition, we created two auxiliary datasets, not explicitly considered in the paper. One dataset was created using a single Active Reading strategy, to analyze the role of strategy diversity, while the second was a mixture of equal proportions of paraphrasing, QA and Active Reading synthetic data, to evaluate multi-strategy training.

III. RESULTS AND ANALYSIS

A. SimpleQA Results

We evaluated the fine-tuned models on the SimpleQA dataset on 100 questions same across all tasks. The evaluation highlights a clear distinction between the parametric-only memory and the Gold Context. For all the tested strategies, the model’s ability to recall facts increases significantly when the text containing the answer for the given question is accessible during the response time. This is illustrated in Fig. 3 by the orange bars. This gap in performance is primarily attributed to the computational resource constraints under the models were trained, limiting the scale of synthetic data generation used for each strategy.

Within this benchmark, the Active Reading strategy demonstrates clear superiority achieving 31 correct answers in the SimpleQA task. This performance is nearly double that of the Paraphrase and Repetition methods, both of which achieved a score of 17 correct answers. These results demonstrate that the Active Reading strategy enables the model to internalize the provided knowledge more efficiently than traditional passive training methods. Regarding the reduced performance of the Question-Answering strategy compared to the results reported in [1], this discrepancy stems from the specific fine-tuning procedure used. While in the original paper the fine-tuning configuration is not explicitly detailed, our implementation focused on next-token-prediction where the model was provided with both the question and the answer during training.

We also evaluated the model trained with the Mixture dataset which, as previously detailed, comprises equal portions of the Paraphrase, Question-Answering and Active Reading datasets. The results indicated that only 8 out of 100 queries were answered correctly in the SimpleQA task, achieving the same performance as the Qwen3-4B-Instruct-2507 baseline. This suggests that the mere aggregation of diverse training strategies does not necessarily yield an additive effect on factual recall within the model’s internal weights, especially considering the strict computational resource constraints under which the models were trained.

B. FinanceBench Results

The evaluation on FinanceBench underscores the inherent difficulty of recalling numerical data from memory. We evaluated 23 distinct queries and similar to the SimpleQA results, the Gold Context remains a critical factor for success as depicted in Fig. 4. By providing the model with the specific context containing the answer, the model is able to better recall and extract facts that otherwise be significantly tougher for the model to retrieve through its internal weights alone. This

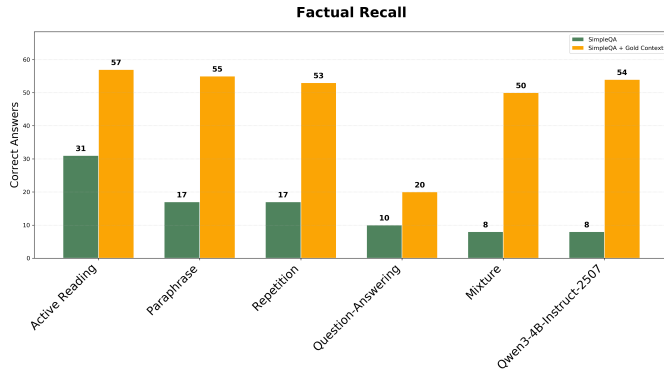


Fig. 3. Comparative analysis of factual recall across 100 discrete queries on the SimpleQA dataset. Green bars denote performance in the SimpleQA task, while orange bars represent performance in SimpleQA task supplemented with Gold Context.

is particularly evident in the performance of the Qwen3-4B-Instruct-2507 model, which was able to increase its answering ability from 3 correct to 22 out of 23 questions in total.

A markedly different result was observed for the Mixture dataset, where the model failed to produce any correct answers when provided with the Gold Context. This unpredicted collapse led us to investigate the causes of such a performance drop. Our analysis of the model’s outputs revealed that in almost every numeric answer the model gave, its response was approximately 1,000, different from the expected answer. Upon further investigation, we found that the Mixture training set itself did not contain a disproportionate frequency of the number 1,000, so simple label bias was not the case. We ultimately attributed this phenomenon to a lack of diverse synthetic data and the small portion of each different strategy that we combined to create the Mixture dataset. These likely resulted in the model falling into a catastrophic interference, where it failed to recall precise numerical values.

Another key difference compared to the results reported by Jessy Lin et al. [1] is that our Active Reading implementation yielded significantly lower scores, underperforming even the base model in the SimpleQA task. While the original study suggests that Active Reading provides robust knowledge internalization, our findings indicate that its efficacy is highly sensitive to the scale of synthetic data. Qualitative analysis of the model’s outputs revealed that instead of starting the response directly with the answer to the given question, the model shifted towards a Chain-of-Thought reasoning style response, often resulting in a failure to meet the response length constraint.

C. Base vs Instruct Model

Given that our experiments use the instruct model, we compare the performance of the base and instruct variants to examine the effect of instruction tuning on factual recall. The evaluation is performed on SimpleQA under both parametric-only memory and Gold Context settings.

As shown in Figure 5, the base model achieves very low performance (2 correct answers) in the parametric-only

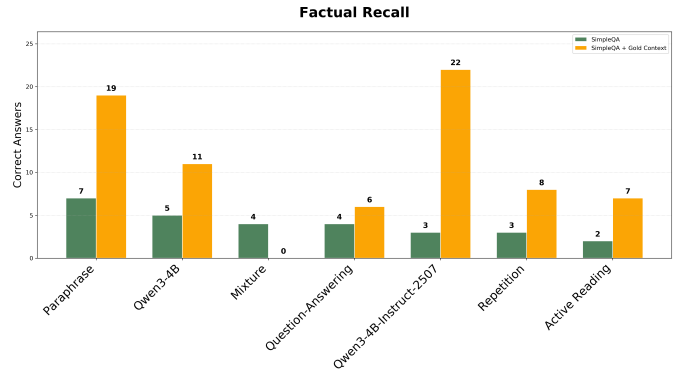


Fig. 4. Comparative analysis of factual recall across 23 discrete queries on the FinanceBench dataset. Green bars denote performance in the SimpleQA task, while orange bars represent performance in SimpleQA task supplemented with Gold Context.

memory setting, reflecting limited parametric memorization of the required factual information. The instruct model improves to 8 correct answers in the same setting, but overall performance remains low. The minimal improvement indicates that instruction tuning does not significantly increase parametric factual knowledge.

In the Gold Context setting, where the required evidence is provided, the performance gap between the two variants is more noticeable. The instruct model achieves 54 correct answers compared to 15 for the base model. This gain demonstrates that instruction tuning improves the model’s ability to follow task instructions and effectively use contextual information.

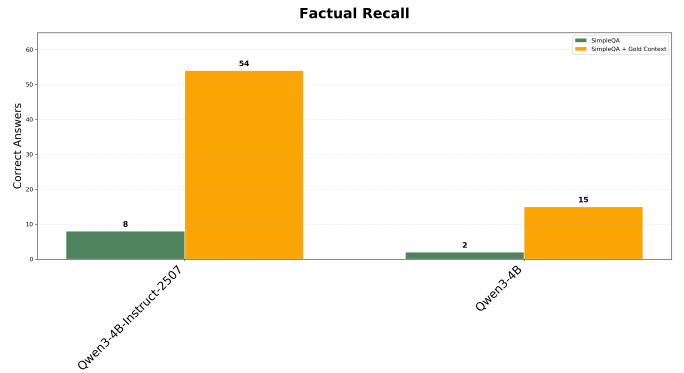


Fig. 5. Performance of the base and instruct variants on SimpleQA under parametric-only memory and gold-context settings. Green bars denote performance on the SimpleQA task, while orange bars represent performance with Gold Context input.

D. Single Strategy Active Reading

To examine the impact of the number of strategies on the effectiveness of Active Reading, we evaluate a restricted variant in which only a single learning strategy is applied per document. We select the first strategy proposed by the model for each document and perform fine-tuning exclusively on the synthetic data generated from this strategy.

As shown in Figure 6, restricting Active Reading to a single strategy per document results in substantial performance degradation across both the parametric-only memory and Gold Context settings. In the parametric-only memory setting, factual recall decreases from 31 to 13 correct answers (−58%). A similar decline is observed in the Gold Context setting, where performance drops from 57 to 25 correct answers (−56%).

The reduced performance can be explained by the limited diversity of training data generated by a single strategy. Because each strategy emphasizes only a specific aspect of the document, it produces synthetic data with limited variation. In contrast, applying multiple strategies extracts information along different conceptual axes, leading to increased data diversity. These results suggest that Active Reading becomes more effective as strategic diversity increases.

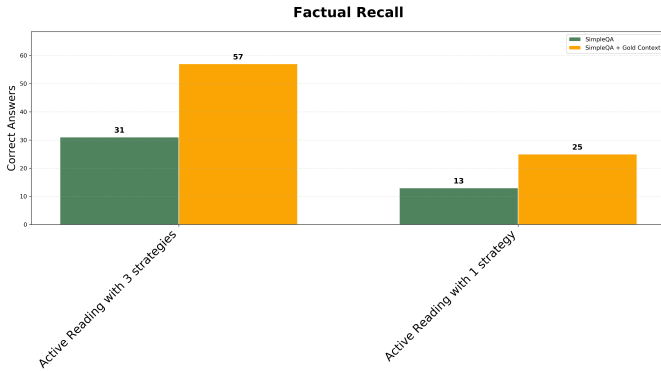


Fig. 6. Performance comparison between the three strategy and single strategy Active Reading. Green bars denote performance on the SimpleQA task, while orange bars represent performance with Gold Context input.

E. Data Diversity

To assess the quality of the synthetic data—and, more specifically, to verify whether the answers to evaluation questions appear verbatim in the training corpus—we evaluate the answerability of each question using a controlled prompting setup. Concretely, we prompt the model with the corresponding context passage, the question, the ground-truth answer, and a proposed answer, and ask it to determine whether the question can be answered solely based on the provided context. The exact prompt used for this classification task is shown in Figure 7.

We apply this procedure to 100 question-answer pairs sampled from the SimpleQA benchmark. The results show that a substantial fraction of the evaluation questions cannot be answered using the training corpus alone. This indicates that the training data does not trivially contain the evaluation answers. Notably, even when considering the repetition (raw) data, the model is still unable to recover the correct answers for many questions. We identify two primary contributing factors: first, the data scraper failed to capture Wikipedia bullet points, which contained many of the relevant answers; second, in some cases, Qwen was unable to detect the presence of an answer even when it existed within the corpus. Nevertheless,

Given the following context, can you answer the question?

```
<context>
{chunk}
</context>
```

```
<question>
{question}
</question>
```

```
<ground truth>
{ground truth}
</ground truth>
```

```
<proposed-answer>
{answer}
</proposed-answer>
```

Can this question be answered from the provided context? Answer **YES** or **NO**.

Fig. 7. Context-answerability classification prompt.

a large proportion of questions remain unanswerable, and this issue becomes more pronounced as the data deviates further from the original corpus, when using strategies like QA or mixture, for example.

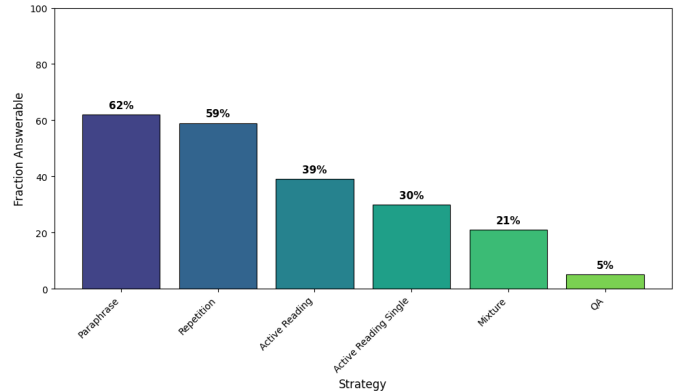


Fig. 8. Fraction of SimpleQA questions that can be answered from the training corpus context. Results are shown for both synthetic and repetition (raw) data.

To evaluate diversity across generated text outputs, we adopted an intra-document protocol partially aligned with the Active Reading paper, under resource constraints. Because the fine-tuning datasets are strongly imbalanced (QA contains many short outputs per document, whereas Active Reading and Paraphrase contain fewer but much longer outputs), we constructed a length-controlled evaluation set. Specifically, we restricted the analysis to the intersection of source documents shared across methods, sampled the maximum feasible number of outputs per document ($k = 3$), and evaluated each output under a fixed word budget T using a deterministic random-window truncation from the full output. Additionally, for QA

we performed repeated subsampling (50 repeats) to account for its substantially larger pool of candidate outputs per document and to reduce dependence on a single arbitrary selection.

As a lexical diversity metric, we used Self-BLEU-4, a self-similarity adaptation of BLEU in which each generated output is treated as the candidate and the remaining outputs for the same document act as references [5]. BLEU scores are averaged across candidates, so lower Self-BLEU indicates lower n -gram overlap among outputs and thus higher lexical diversity. In the three-way comparison (Active Reading, Paraphrase, QA), QA exhibited the highest Self-BLEU (least diverse), consistent with its more formulaic short-form generation. Concretely, for both $T = 128$ and $T = 256$, QA scored 0.079, whereas Paraphrase and Active Reading achieved substantially lower values (below 0.033 in our matched setting).

Having established that QA is the least diverse under this lexical metric, we conducted a secondary analysis comparing only Active Reading and Paraphrase, enabling a larger matched set of source documents and a higher budget up to $T = 512$. The two methods appear approximately tied in lexical diversity, with only small differences across T (maximum absolute difference ≈ 0.003), and neither method dominates consistently. This outcome is not inconsistent with the paper’s diversity analysis, which relies on multi-generation scaling whereas our study is post-hoc over pre-generated outputs with fixed k . Finally, because Self-BLEU is based on n -gram overlap, structured templating in Active Reading outputs can increase overlap and inflate Self-BLEU, even when outputs differ in content. Thus, Self-BLEU should be interpreted primarily as a measure of lexical (not semantic) diversity.

IV. LIMITATIONS AND FUTURE WORK

We faced significant limitations in our experiments, mainly because of the large gap between the resources used in the Active Reading paper and ours. Our main challenge was to apply the framework accurately enough for a proof of concept while accounting for limited resources and restricted data access. We had to adjust the data-generation step to produce a sufficient number of tokens within a short time while ensuring a balanced token production across the methods for a safe comparison.

To optimize computational efficiency, we employed a 4-bit quantized model using larger text chunks on a corpus subset, and adopted QLoRA parameter-efficient fine-tuning instead of full-scale fine-tuning. The use of QLoRA instead of Full Fine-Tuning (Full-FT) introduces distinct mechanical differences in how the model learns, which can influence the results of knowledge-intensive tasks. While QLoRA imposes intrinsic constraints on the representational capacity for new knowledge, it simultaneously functions as a regularizer, mitigating common Full-FT issues such as overfitting and catastrophic forgetting. Furthermore, model quantization can also lead to a slight degradation of the models’ retrieval abilities, thereby affecting our results.

Another limitation concerns the Active Reading configuration itself, as our replication implemented only the task-agnostic variant due to limited time and computational resources, omitting the task specific formulation proposed in the original work. Additionally, the FinanceBench evaluation relied on a 23-question subset provided by the Hugging Face dataset; while methodologically consistent with the available data this restricted sample size weakens statistical significance and limits comparability with the original large scale benchmark setting.

Despite the aforementioned constraints, our findings suggest that the Active Reading method maintains fundamental robustness, indicating that its efficacy is primarily contingent upon data quality rather than architectural scale alone. Future research should prioritize exploring the scaling impacts, analyze the effectiveness of each strategy and the domain-specific effectiveness of each strategy. First, investigating the scaling laws specifically for QLoRA is crucial; while the original authors demonstrated linear performance gains with full fine-tuning up to 4 billion words, it remains to be seen if parameter-efficient methods exhibit similar endurance or if they saturate earlier due to limited capacity. Second, conducting a granular ablation study on individual learning strategies could identify the most efficient strategies, enabling optimized training for smaller models. Furthermore, this analysis should extend to domain-specific effectiveness, investigating which strategies yield the highest recall for particular fields—such as determining if structural strategies like ‘timelines’ favor historical contexts, while reasoning-heavy strategies excel in financial or medical domains. Finally, another path could be an exploration on Reasoning- Augmented Active Reading, combining factual learning with reasoning oriented prompts, thereby building a model with functional knowledge application abilities.

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